

TCS34725 Colour Calibration — Capture Worksheet

docs/colour-calibration.md

Recording sheet for fitting a colour-correction matrix (CCM)

Use real physical LEGO bricks — not this printed page. Printer ink/toner has a different reflectance spectrum than genuine ABS plastic, so printed swatches would fit the matrix to your printer, not to the bricks the sensor actually reads. This sheet is a **recording sheet**: point the sensor at real bricks and write the numbers down here.

Setup checklist (do these once, before recording anything)

1. Use one consistent light source for the whole session — avoid mixing daylight with room lighting, and avoid direct sunlight (it shifts colour temperature as it moves).
2. Fix the distance and angle from the sensor to the brick and keep it the same for every sample — a spacer/jig helps more than trying to hold it steady by hand.
3. Don't change **poll_ms** during the session — TCS34725's gain/integration are derived from it, so switching rates mid-session mixes samples from different effective gains.
4. Run **white-balance Calibrate** against a real white brick first, before recording any samples below.
5. Read the sensor in **col_rgb255** or **col_full** mode and record the **r, g, b** numbers from the dashboard's "rgb" line for each sample.

Which bricks to use

Use genuinely **solid, saturated** official LEGO colours — not pastel, translucent, glow, metallic, or speckled pieces. A good matrix fit needs well-separated, fully-saturated primaries; a washed-out or translucent piece gives the fit less to work with and can make it worse rather than better.

Essential samples (minimum needed for a usable fit)

Colour	Brick used (note the part/colour name)	r, g, b		
White		r:	g:	b:
Black		r:	g:	b:
Red		r:	g:	b:
Green		r:	g:	b:
Blue		r:	g:	b:
Yellow		r:	g:	b:

Optional — for a fuller 12-colour fit later

Colour	Brick used (note the part/colour name)	r, g, b		
Violet		r:	g:	b:
Light blue / azure		r:	g:	b:
Orange		r:	g:	b:
Purple		r:	g:	b:
Cyan / teal		r:	g:	b:
Silver		r:	g:	b:

Once filled in, send the numbers back to fit TCS_CCM/TCS_CCM_BIAS in firmware/components/sensor/sensor_transform.c. See docs/colour-calibration.md for background on why a single white-point calibration can't fix this class of error.